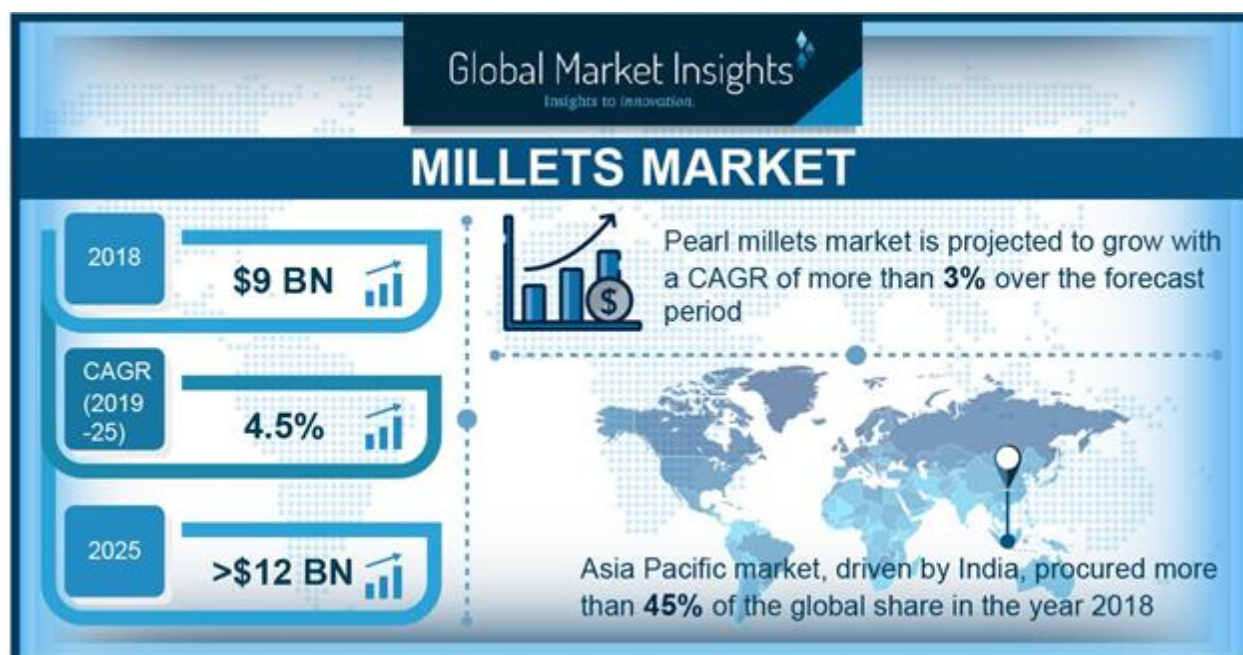




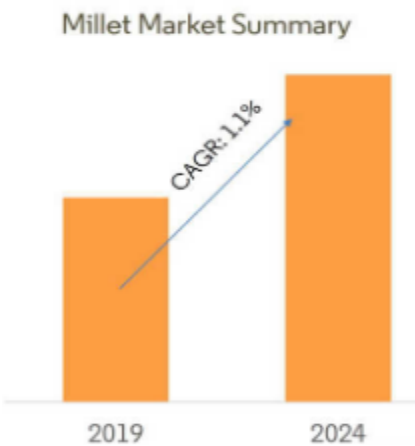
## Market Intelligence Report for millets

### 2023 Will be the International Year of Millets

Millet is a common term to categorize small-seeded grasses that are often termed nutri-cereals or dryland-cereals, and includes sorghum, pearl millet, ragi, small millet, foxtail millet, proso millet, barnyard millet, kodo millet and other millets. Millets are the cereal crops generally small-seeded and known for high nutritive value. Increasing interest in reviving the consumption of millets across various countries is favouring the growth prospects of this market in recent years. A number of initiatives are also being undertaken toward enhancing millet cultivation and consumption to reduce health risks caused due to diabetes, obesity, cardiovascular diseases. Unified approaches among millet suppliers and food and beverage manufacturing companies to fulfil the demand of millets has been crucial to proliferate its industry size.



With an annual global output of 25 million tons, millet has been one of the basic nutrients of humans for 4 thousand years in Africa and Asia and for Europe until the end of the Middle Age. The global millet consumption has declined at a rate of 0.9% and expected to witness positive movement during 2019-2024.

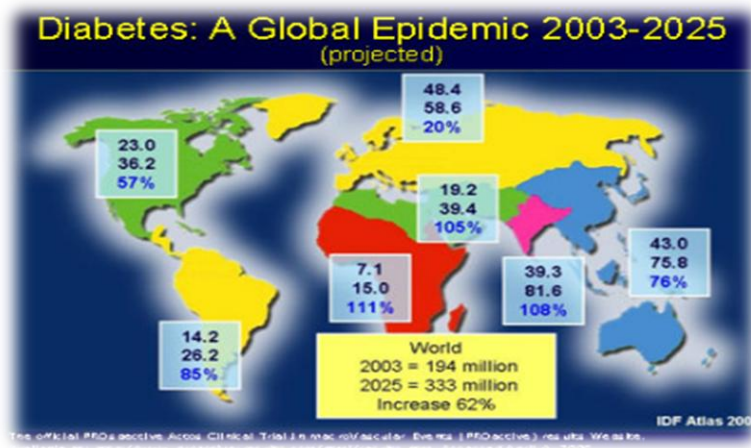


“The millets market is set to grow from its current market value of more than \$9 billion to over \$12 billion by 2025. Favorable government initiatives to proliferate the global millets market size over 2019-2025”.

## What would drive Millet trade in 2025?

|                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Growing inclination of urban population towards healthy food in Asia Pacific                                                                                                                                                                                                              | Millets contains calcium, iron and fibers which helps to fortify essential nutrients for the healthy growth in children. The usage of millets in infant food and nutrition products is increasing and many manufacturers are expanding their business |                                                                                                                                                                                                                                                                         |
| Increasing unsustainable nature of rice and wheat production which are water intensive and are likely to be unsustainable, as freshwater resources are depleting around the globe. Millet grows easily in dry climate, have smaller harvesting period and require minimal water quantity. | In terms of nutritional property, they are superior to certain highly consumed cereals such as rice and wheat.<br>High product prices in comparison with largely consumed grains are acting as a hindrance for penetration in urban food market.      | Growing number of diseases such as diabetes, obesity and cardiovascular problems such as heart attack, coronary artery disease, arrhythmias etc. Millets have high content of proteins and minerals such as calcium, iron etc. that can help in avoiding such diseases. |
| Photo-insensitive & resilient to climate change, millets are hardy, resilient crops that have a low carbon and water footprint, can withstand high temperatures and grow on poor soils with little or no external inputs.                                                                 |                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                         |

The urban lifestyle choices and related food habits have given rise to number of diseases such as diabetes, obesity and cardiovascular problems such as heart attack, coronary artery disease, arrhythmias etc.

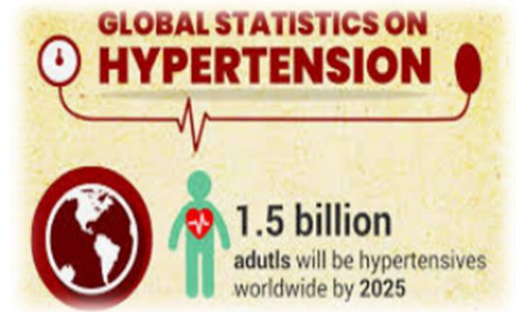


“Millets are small-seeded cereals that retain excellent nutritional qualities; superior to certain highly consumed cereals such as rice and wheat. Millets have high content of proteins and minerals such as calcium, iron etc. that can help in avoiding such diseases. **Therefore, growing awareness amongst population regarding health benefits associated with millets consumption will boost industry growth by 2025”.**

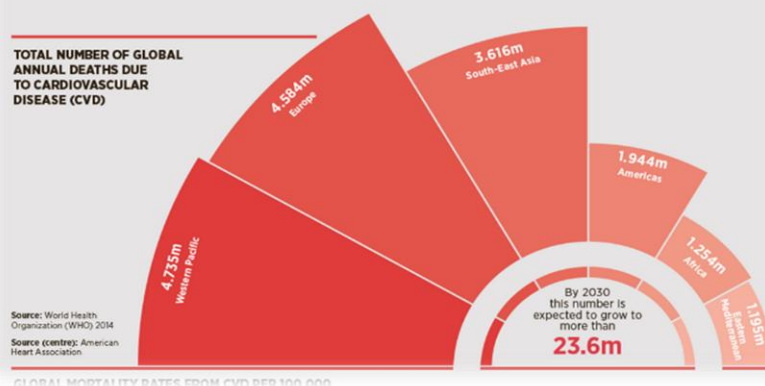


#### 5. Estimated prevalence of severe obesity by 2025 - adults

On current trends, 177 million adults worldwide will be severely obese and in need of treatment by 2025 – up from 98m in 2014.



## Global cardiovascular disease outlook



**“By 2030, number of patients with cardiovascular disease to grow more than 23.6 million giving good boost to millet consumption”.**

It also contains calcium, iron and fibers which help to fortify essential nutrients for the healthy growth in children. Millet has more protein than rice and is rich in vitamins A and B, iron, phosphorus, magnesium and manganese. **The usage of millets in infant food and nutrition products is increasing and many manufacturers are expanding their business operations by acquiring smaller firms.** It has also found usage in beverages like beer. Gluten-free beers are specially produced for individuals focused on reducing their gluten intake or diagnosed with celiac disease or a gluten intolerance. **Breweries have started to add gluten-free beer option to increase their consumer base.** Numerous developing nations and federal governments are framing long-term policies to roll out nutrition programs aimed at tackling malnutrition. The support of government initiatives and inclusion of millets in various food and beverage products are likely to enhance its industry size.

Millets have a limited shelf life. This factor is estimated to hamper the market growth. However, the gluten-free property is expected to provide lucrative opportunities for the production of gluten-free and low GI food products. Millets, buckwheat, and quinoa are widely used in preparing popular dishes such as waffles, pasta, pizza, and sandwiches. Changing outlook towards low cholesterol and fat free alternatives, and rising awareness towards healthy diets may boost gluten free food demand. Increasing preference for healthy bakery preparations among consumer is motivating manufacturers to use natural and gluten free ingredients. Gluten free food products offer same taste & consistency without high concentration of saturated and trans-fat. Rising inclination to adopt healthy food habits and shift towards low calorie alternatives may stimulate market share.

The demand for gluten free food products is increasing across the globe due to their benefits such as healthier lipid profile and reduces risk of heart diseases

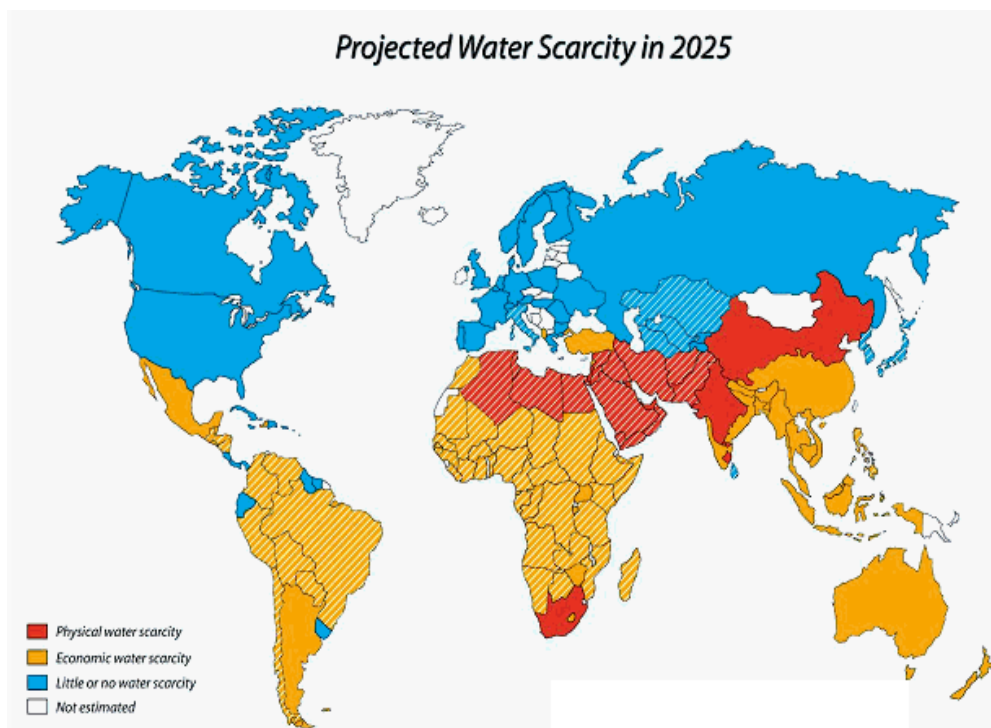
- Increasing prevalence of digestive disorders in Europe market may trigger the demand for light and gluten free alternatives.
- Changing consumer perceptions towards the use of natural ingredients and increasing demand for high protein grains such as oat, buckwheat, and quinoa is expected to provide opportunities for industry expansion.
- Companies are manufacturing products by sourcing organic ingredients and investing in alternative manufacturing solutions.
- Stringent regulations in North America & European market towards labelling of gluten free food will increase consumer awareness related to the products.



## However there will also be challenges

Lack of product recognition by various government agencies is expected to be a major challenge for business expansion in the projected time period. Further, *use of the product for treating chronic diseases is not recognized by government agencies such as the FDA*. Stringent government regulations towards drug usage owing to rising cases of chronic illness is likely to hamper coenzyme industry growth.

As per the UN Food and Agriculture Organization's data, agriculture accounts for 70% of total water consumption among these sectors. It is highest for Asia and Africa where agriculture is in primary sector of economy. Among agricultural crops, rice and wheat are staple food in large parts of globe. However, they are water intensive and are likely to be unsustainable, as freshwater resources are depleting around the globe. Millet grows easily in dry climate, have smaller harvesting period and require minimal water quantity. They can be a sustainable alternative to rice and wheat, as a new staple food. It can also help in providing food security to large population in the coming years.



**“Water withdrawals are predicted to increase by 50 percent by 2025 in developing countries, and 18 per cent in developed countries. In 60 percent of European cities with more than 100,000 people, groundwater is being used at a faster rate than it can be replenished”.Millet grows easily in dry climate, have smaller harvesting period and require minimal water quantity.**

With steadily increasing demand from urban population in Asia Pacific, mainly India and China, farmers have begun drifting towards millets cultivation over rice and wheat. High product prices in comparison with largely consumed grains are acting as a hindrance for penetration in urban food market. These high prices are owing to lower production volume and weak supply chain in millets market as compared to rice and wheat.

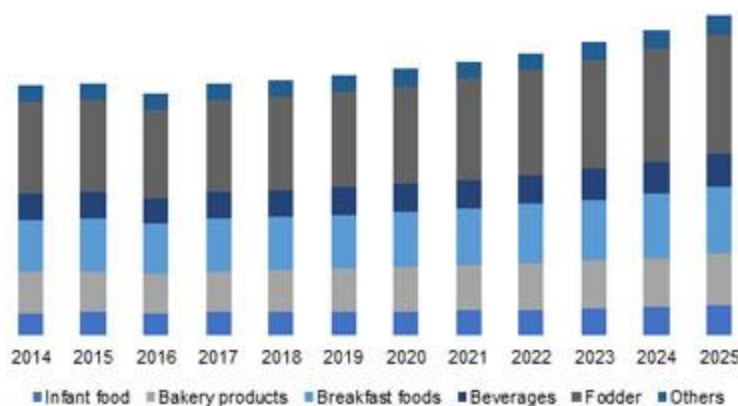
Photo-insensitive & resilient to climate change, millets are hardy, resilient crops that have a low carbon and water footprint, can withstand high temperatures and grow on poor soils with little or no external inputs.

## Target Markets for Millets in 2020

Despite their prominent nutritional qualities, millets have received less attention than the major cereals. Millets play an important role in food security and economy of many countries in Africa and Asia. They are gaining prominence in Europe and North America due to their gluten-free and hypoglycemic properties. Millets maintains its importance in human nutrition in the hot and arid countries of Africa and Asia, although it is cultivated as a feed plant in Turkey, in Western Europe and in the United States.

Although millet represents less than 2 percent of world cereal utilization, it is an important staple in a large number of countries in the semi-arid tropics, where low precipitation and poor soils limit the cultivation of other major food crops.

Millet utilization is mostly confined to the developing countries, even more so after production and utilization fell sharply in the CIS, the largest producer in the developed world. Accurate data are not available for most countries, but it is estimated that about 80 percent of the world's millet (and over 95 percent in Asia and Africa) is used as food, the remainder being divided between feed (7 percent), other uses (seed, beer, etc.,) and waste. A large volume of millet produced, is locally grown and consumed. The remaining produce is used to make beer, infant food and breakfast. It is commonly used to make breakfast products such as flakes and cereals. Millet beer is popular in African culture and is also finding a market in countries such as India and the U.S.



**Milletts are gaining a significant momentum in feedstock and food purposes, on the back of its high productivity in dry and high-temperature conditions. Revenues from sales of millets across the globe is forecast to increase from over US\$ 10,000 Mn in 2017 to exceed US\$ 13,500 Mn by 2022-end.**

Milletts based products such as flour, flakes, cookies, etc. are more than ever increasingly visible in consumer market. The high protein content of these grains makes them ideal for vegetarian and vegan population, largely based in the U.S., Europe and Asia Pacific. Breakfast cereals such as flakes and local recipes have seen an ascending trend of adoption by African and Asian population Thus, millets based breakfast foods generated revenue over USD 2 billion in 2018 and likely to witness maximum gains due to increasing pattern of fibrous and gluten free food consumption.

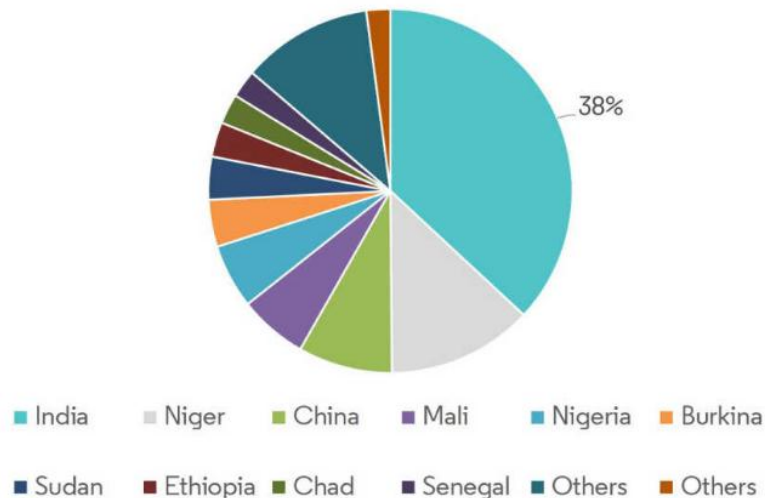
Milletts based infant food such as porridge are ideal for infant growth and will help to lower the occurrence of malnutrition in infants and babies. Bakery products such as packaged cookies are slowly Gaining prominence owing to its easy availability in availability throughout supermarkets and e-commerce sites, especially in Asia Pacific. Millet Beer is a traditional part of African culture and ventures related to millet beer production are also finding place in Asia Pacific, Europe and North American beverage market. In the U.S., the product is also commonly used as fodder for animals and birds.

Lignans, an essential phytonutrient present in millet, help reduce the risk of heart diseases. Pearl millets are rich in insoluble fibre and aid in better digestion, and are known for their anti-cancer properties. Foxtail millets are not just rich in magnesium that assists to regulate blood pressure levels, they are also high in iron and calcium and help boost immunity levels.

## Africa dominates global consumption in 2020

More than 40% of global millet consumption is held by African countries mainly Niger, Mali, Nigeria, Burkina, and Sudan. Millets are popular in developing regions, like India and Africa, where food and nutritional security are the major challenges. The high drought tolerance capacity of the crop allows it to be cultivated in the adverse environmental conditions of the regions. India is the world's leading producer of millet. Public and private investments are limited to millet seed development and production. International prices for millets are highly volatile, determined largely by supply volumes, and are usually unrelated to those of other major coarse grains, such as maize, sorghum, or barley.

Millet Market: Major Consumers, By Volume in %,



Per caput food consumption of millet varies greatly between countries; it is highest in Africa, where millet is a key food staple in the drier regions. Millet represents about 75 percent of total cereal food consumption in Niger and over 30 percent in most other countries in the Sahel. It is also important in Namibia (25 percent of total cereal food consumption) and Uganda (20 percent).

Outside Africa, millet food consumption is important in parts of India, China and Myanmar. Utilization is negligible in Latin America, the Caribbean and all the developed countries. The exception was the former USSR until the late 1980s. However, under the economic transition process and the removal of price subsidies, utilization in the USSR/CIS fell sharply



Millet is a high-energy, nutritious food, especially recommended for children, convalescents and the elderly. Several food preparations are made from millet, which differ between countries and even between different parts of a country. These consist primarily of porridge or pancake-like flat bread. However, because wholemeal quickly goes rancid, millet flour (prepared by pounding or milling) can be stored only for short periods.

#### Animal feed

Utilization of millet grain as animal feed is not significant. It is estimated that less than 2 million tons, (about 7 percent of total utilization), is fed to animals, compared with about 30 million tons of sorghum (almost half of total output). In the developing countries, use of millet grain for animal feed is concentrated in Asia; very little is fed in Africa. However, millet fodder and stover are a valuable and critical resource in the crop/livestock systems where millet is grown.

Feed use estimates are heavily influenced by assumptions made for China, the world's third largest producer. In fact, little reliable information is available on feed use in this country. Based on very rough calculations of feed use in the CIS, it is estimated that about 1.0 million tons per annum are currently used as animal feed in the developed countries. Western Europe, North America and Japan together use slightly over 200,000 tons, almost exclusively as bird seed. Recent increases in the use of pearl millet as a lower-cost substitute for maize feed in aquaculture and dairy and poultry farming in India and the southeastern United States are not well documented, and, in any case, at the moment represent only a small fraction of overall feed grain. There are few other uses of millet. Small quantities of finger millet are used in Zimbabwe for commercial brewing and opaque beer utilization.

## Millet Type: Focus Areas for 2025

The millets market can be segmented based on type into:

- Pearl millet
- Finger millet
- Proso millet
- Foxtail millet
- Others (Japanese barnyard millet, Kodo millet, etc.)

**Pearl millets accounted for the largest share in millets markets in 2018. The product will be growing with more than 3% CAGR by 2025.**

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In 2018, pearl held highest share in terms of product volume in millets market and is likely to expand at more than 3% CAGR by 2025. Massive demand share of pearl millets is due to its rich nutritional value as it has 8 times higher iron content than rice. Finger millets are popular among several cultures and have demand from processors. Foxtail, proso and other millets are also rich in fiber, calcium and other minerals. Increasing fibrous food demand for breakfast meals in shall positively impact on the industry growth over the forecast timeframe. Thus, breakfast food accounted for over than 20% of the overall revenue generation for millets market in 2018.

The increased share of pearl millet is related to the cultivation and consumption pattern of the region it is grown in. **Organic millets form an integral part of organic food products bought by consumers who prefer healthy food options.** The growing section of such kind of consumer in urban areas of North America, Europe and Asia Pacific will form a lucrative segment for such organic products in the coming years.

- Pearl millet adapts well to hot climates and is also resistant to drought conditions. It grows well compared to any other cereal in hot and dry regions. Pearl millet yields well on sandy soils where most other crops fail. Pearl millet is the major millet grown in Africa. It is also the fourth most significant food crop/cereal grown in India.
- Proso millet is primarily suitable to a dry continental climate than most other millets. Also known as the common millet, it exhibits the lowest water necessity. Proso millet is predominantly cultivated in Eastern Asia, including Japan, India, Mongolia, and Manchuria; and eastern and central Russia.
- Kodo millet is grown commercially in India; however, it is an extensive tropical weed. This cereal crop is often grown on poor soils and is drought resistant.

## ASIA PACIFIC, MIDDLE EAST AND EU WOULD REMAIN TOP IMPORTERS OF MILLETS IN 2020

|                           | Value imported in 2018 (USD thousand) | Trade balance in 2018 (USD thousand) | Quantity imported in 2018 | Unit value (USD/unit) | Average tariff (estimated) applied by the country (%) |
|---------------------------|---------------------------------------|--------------------------------------|---------------------------|-----------------------|-------------------------------------------------------|
| Indonesia                 | 18,547                                | -18,539                              | 54,858                    | 338                   | 4.7                                                   |
| Germany                   | 13,041                                | -10,040                              | 26,618                    | 490                   | 4.2                                                   |
| Belgium                   | 8,529                                 | -6,596                               | 19,787                    | 431                   | 4.2                                                   |
| Korea, Republic of        | 7,572                                 | -7,552                               | 12,716                    | 595                   | 1.8                                                   |
| Netherlands               | 5,913                                 | -2,927                               | 14,829                    | 399                   | 4.2                                                   |
| Canada                    | 4,962                                 | -4,196                               | 12,713                    | 390                   | 0                                                     |
| United Arab Emirates      | 4,906                                 | -4,493                               | 16,953                    | 289                   | 0                                                     |
| Japan                     | 3,563                                 | -3,563                               | 8,421                     | 423                   | 0                                                     |
| United Kingdom            | 3,553                                 | -3,508                               | 7,064                     | 503                   | 4.2                                                   |
| Italy                     | 3,539                                 | -3,298                               | 9,786                     | 362                   | 4.2                                                   |
| Thailand                  | 2,682                                 | -2,631                               | 6,795                     | 395                   | 20.2                                                  |
| Saudi Arabia              | 2,366                                 | -2,366                               | 6,388                     | 370                   | 0                                                     |
| Spain                     | 2,361                                 | -2,061                               | 7,050                     | 335                   | 4.2                                                   |
| Iran, Islamic Republic of | 2,352                                 | -2,238                               | 5,854                     | 402                   | 10                                                    |
| Austria                   | 2,129                                 | 2,131                                | 4,107                     | 518                   | 4.2                                                   |
| Yemen                     | 1,625                                 | -780                                 | 6,605                     | 246                   | 23.2                                                  |
| Switzerland               | 1,585                                 | -1,569                               | 1,844                     | 860                   | 7.1                                                   |
| Pakistan                  | 1,527                                 | -1,221                               | 5,631                     | 271                   | 3                                                     |

## POTENTIAL EXPORT MARKETS FOR INDIAN MILLETS IN 2020



## India's position against competing players in potential destinations

| Indonesia   | Country                  | % Share | CIF   | Duty | FLP   |
|-------------|--------------------------|---------|-------|------|-------|
|             | USA                      | 97.20%  | 336   | 5%   | 353   |
|             | China                    | 2.10%   | 526   | 0%   | 526   |
|             | Ukraine                  | 0.30%   | 321   | 5%   | 337   |
|             | Russia Federation        | 0.30%   | 314   | 5%   | 330   |
|             | India                    | 0.10%   | 348   | 5%   | 365   |
| Austria     | Romania                  | 34.00%  | 292   | 0%   | 292   |
|             | Germany                  | 20.00%  | 1,495 | 0%   | 1,495 |
|             | Ukraine                  | 17.60%  | 1,078 | 0%   | 1,078 |
|             | Hungary                  | 9.40%   | 385   | 0%   | 385   |
|             | Czech Republic           | 7.20%   | 614   | 0%   | 614   |
| Switzerland | Austria                  | 40.40%  | 818   | 9%   | 891   |
|             | Germany                  | 18.60%  | 776   | 9%   | 845   |
|             | France                   | 12.60%  | 926   | 9%   | 1,008 |
|             | Netherlands              | 12.00%  | 2,794 | 9%   | 3,043 |
|             | Hungary                  | 5.90%   | 461   | 9%   | 502   |
| Philippines | United States of America | 90.90%  | 336   | 0%   | 336   |
|             | China                    | 3.80%   | 351   | 0%   | 351   |
|             | Ukraine                  | 2.70%   | 322   | 0%   | 322   |
|             | Canada                   | 1.70%   | 289   | 0%   | 289   |
|             | Latvia                   | 0.90%   | 310   | 0%   | 310   |
| New Zealand | USA                      | 53.00%  | 551   | 20%  | 661   |
|             | China                    | 24.60%  | 1,065 | 18%  | 1,257 |
|             | India                    | 22.50%  | 421   | 20%  | 505   |
|             | Australia                | 0.30%   | -     | 20%  | -     |